

10 Disrupting the Dominant Paradigm

This book has described a human-inspired approach to modeling cognitive agents that understand their own behavior and, therefore, can explain it in normal, human terms. Any cognitive system developed using this modeling strategy is, in our parlance, a LEIA.

The fact that LEIAs are being designed to collaborate with people gives rise to certain priorities, including meaning-oriented language processing, human-oriented explanation, and lifelong learning. It also opens up certain opportunities, including distributing tasks across agents and people in a way that plays to their respective strengths and enabling agents to learn interactively, as human apprentices would. Since LEIA development is task-oriented rather than method-oriented, it accommodates a variety of metalanguage formalisms, algorithmic solutions, and means of acquiring knowledge, including taking advantage of the capabilities of large language models (LLMs).

Any specific LEIA development effort requires a host of technical and methodological decisions, and our group's status as a university laboratory naturally affects such decision-making. For example, we prioritize scientific discovery over resource acquisition, we attempt to solve problems in a manner that establishes a theme for which technologists can work out the variations, and we develop demonstration systems as deliverables for funded research.

The title of this book was selected judiciously. Achieving trustworthy AI *will* be a long game, and it would be a mistake for society to entrust certain kinds of responsibilities to systems that are not operating with intentional intelligence. Key to this statement is the modifier: *certain kinds of* responsibilities. Machine-learning-based AI systems produce useful results, including some that contribute to LEIA operation. However, in our estimation,

machine learning is not the whole answer to achieving humanlike artificial intelligence. This is the nontraditional claim that renders the LEIA program of R&D disruptive.

In the popular press, there is a concerning blurring of the line between scientific reality and sci-fi fantasy. The popular imagination is naturally animated by the singularity scenario, the idea of machines becoming Terminators, and the notion that machines could become so fixated on accomplishing a task—like making paper clips in Nick Bostrom’s thought experiment—that they would pursue it to the point of harming humanity.¹ Although such imaginings make for lively dinner conversation, they do not realistically predict the trajectory of AI development.

Similarly problematic is the overreliance on metaphors both within the AI community and as a tool for explaining AI to non-specialists. Metaphors can help scientists think about complex problems, but they can also become so entrenched as to inhibit creative thought. They can provide the public with insights into scientific discoveries, but they can also create misunderstandings due to the public’s inability to recognize the distance between the metaphor and the reality it seeks to evoke. The professional and popular literature on AI is replete with metaphors, starting with the term *artificial intelligence* itself and proceeding to such things as *neural nets*, *machine learning*, *machine reasoning*, *computer vision*, *chat bots*, and *adversarial AI*. The metaphorical links of AI terminology to human experience inevitably contribute to unrealistic assessments by non-specialists of both the power of modern AI and the threats it engenders.

One metaphor that is particularly misleading is that ML-based AI systems learn automatically, which implies without the need for human labor. However, in reality, ML-based AI relies on a massive workforce of human data annotators. Josh Dzieza² paints a stark picture of the work lives of these annotators: their job is difficult, repetitive, and boring, and the workers are given no insight into how their efforts will contribute to any tangible goal. Dzieza explains, “When AI comes for your job, you may not lose it, but it might become more alien, more isolating, more tedious.” Although LEIA R&D also requires human labor, it is of a completely different kind: trained knowledge engineers will enhance interpretable knowledge bases in particular ways for clearly identified purposes. This makes knowledge engineering more akin to the desirable profession of software engineering than to the grueling task of corpus tagging.

It is not only metaphors that mislead; so can anthropomorphizing. As an article in *The Economist* points out, humans have an ancient tendency to anthropomorphize, but it should be resisted when thinking about AI technologies like LLMs. “These systems, including those that seem to converse, merely take input and produce output. At their most basic level, they do nothing more than turn strings like 0010010101001010 into 1011100100100001 based on a set of instructions. Other parts of the software turn those 0s and 1s into words, giving a frightening—but false—sense that there is a ghost in the machine.”³

How AI will play out over time depends on unknowns involving science, technology, and how society’s decision-makers choose to allocate resources. Those decisions, in turn, will involve not only the decision-makers’ priorities and cost-benefit analyses but also their understanding of the different technologies on offer. The latter makes it imperative for the scientific branch of the AI community to explain the different technologies plainly so that real people know what they are getting—and, more importantly, what they are putting their faith in. *The Economist* formulates this idea as follows:

AI is too important for loose language. If entirely avoiding human-like metaphors is all but impossible, writers should offset them, early, with some suitably bloodless phrasing. “An LLM is designed to produce text that reflects patterns found in its vast training data,” or some such explanation, will help readers take any later imagery with due scepticism. Humans have evolved to spot ghosts in machines. Writers should avoid ushering them into that trap. Better to lead them out of it. (*The Economist*)⁴

Since nobody can predict how AI will evolve over the next five years, not to speak of the next twenty or fifty, it would be unwise for society to place all its bets on a single theoretical or methodological approach. LEIAs offer a counterpoint to mainstream, data-driven AI, with the potential to earn an unprecedented, and critically needed, level of trust by the humans responsible for their operation.